EE 115 Homework 7

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Given

$$u_{FM}(t) = A_c \cos\big(2\pi f_c t + \theta(t)\big), \qquad \theta(t) = \theta(0) + 8\pi \int_0^t m(\tau)\, d\tau.$$

Comparing to the standard form $2\pi k_f \int m(\tau)\, d\tau$ shows $k_f = 4$ Hz/unit. Thus the peak frequency deviation is

$$\Delta f = k_f\, |m(t)|_{\max} = 4\, |m(t)|_{\max},$$

and Carson's rule gives the approximate FM bandwidth

$$W \approx 2\big(\Delta f + f_{m,\max}\big),$$

where $f_{m,\max}$ is the highest frequency present in $m(t)$.

1. Bandwidth of $m(t)$ is 10 Hz and $|m(t)|_{\max} = 20$.

$$\Delta f = 4(20) = 80\,\text{Hz}, \qquad W \approx 2(80 + 10) = \boxed{180 \text{ Hz}}.$$

2. $m(t) = 2\cos(10\pi t) = 2\cos(2\pi \cdot 5\, t)$ has $f_{m,\max} = 5$ Hz and $|m(t)|_{\max} = 2$.

$$\Delta f = 4(2) = 8\,\text{Hz}, \quad \beta = \frac{\Delta f}{f_{m,\max}} = \frac{8}{5} = 1.6, \quad W \approx 2(8 + 5) = \boxed{26 \text{ Hz}}.$$

3. $M(f) = \text{rect}(f/3)$ is nonzero for $|f| \le 3/2$, so $f_{m,\max} = 1.5$ Hz. The inverse transform is $m(t) = 3\ \text{sinc}(3t)$, giving $|m(t)|_{\max} = 3$ at $t = 0$.

$$\Delta f = 4(3) = 12\,\text{Hz}, \qquad W \approx 2(12 + 1.5) = \boxed{27 \text{ Hz}}.$$

4. $m(t) = \displaystyle\int_0^t 2\cos(10\pi\tau)\, d\tau = \frac{1}{5\pi}\sin(10\pi t)$ has $f_{m,\max} = 5$ Hz and $|m(t)|_{\max} = \frac{1}{5\pi}$.

$$\Delta f = 4\left(\tfrac{1}{5\pi}\right) = \tfrac{4}{5\pi}\,\text{Hz}, \qquad W \approx 2\big(\tfrac{4}{5\pi} + 5\big) = \boxed{10 + \tfrac{8}{5\pi} \text{ Hz} \approx 10.51 \text{ Hz}}.$$

5. $m(t) = \dfrac{d}{dt}[2\cos(10\pi t)] = -20\pi\sin(10\pi t)$ has $f_{m,\max} = 5$ Hz and $|m(t)|_{\max} = 20\pi$.

$$\Delta f = 4(20\pi) = 80\pi\,\text{Hz}, \qquad W \approx 2(80\pi + 5) = \boxed{160\pi + 10 \text{ Hz} \approx 5.13 \times 10^2 \text{ Hz}}.$$